1. Exact Cosmos model versions and checkpoints (Reason, Transfer, and whether Cosmos 3 was used for detection)

   Cosmos Reason: /Cosmos3-Nano reasoner -> 16 billion total params -> 8 billion for reasoning -> 8 billion for generating. The reasoner worked great. Tried generation and found lack of support for video 2 video transfer.

   Cosmos Transfer: Cosmos Transfer 2.5

   Cosmos Predict: nvidia/Cosmos-Predict2-2B-Video2World

   .

2. Role of Cosmos Curate.

Cosmos curate is a family of multiple models used to split, caption, and turn long clips into embeddings. It is also met to down sample clips to the appropriate fps and time for cosmos reason, transfer and predict. We downsampled everything to 15 seconds and 4 fps. We used transnetv2 (part of the cosmos curate codebase) to clip up our original 8 clips that were each about 10 minutes in length to generate 129 clips.

3. Which segmenter turned boxes into masks, and whether masks were per-frame or tracked.

   This was SAM2. Your Cosmos Transfer Colab had a segmentation notebook where Cosmos Reason produced bounding boxes (bbox_2d coordinates), and then SAM2 turned those boxes into pixel-level masks. Your notebook (Cell 11) output seg_mask.mp4 (binary mask) and seg_overlay.mp4 (overlay visualization). The masks were tracked across frames — SAM2 is a video segmentation model, not per-frame — which is why you got temporally consistent masks for control map generation. You also had a subtraction approach: detect the surface plus objects on it, then subtract to isolate what to change.

4. Control weights, guidance scale, steps, resolution, frames, seeds.

python

```python
"edge":  {"control_weight": 0.556},
"vis":   {"control_weight": 0.028},
"depth": {"control_weight": 0.222},
"seg":   {"control_weight": 0.194},
```

As for what you found during tuning — edge dominates at ~56% of the total budget because it's doing the heavy lifting: preserving object shape and boundaries. Pushing edge any lower caused objects to lose coherent shape entirely. Depth sits at ~22%, and increasing it further flattened the scene onto a 2D plane — the model would collapse the depth structure instead of preserving it. That left vis and seg as the two knobs you actually had room to experiment with, and they share the remaining ~22% between them. Vis stays near zero (~3%) because any higher and it fights the color/texture changes you're trying to make — it locks in the original appearance. Seg at ~19% provides just enough mask-based guidance to keep the generation spatially grounded without overconstraining it.

The takeaway for your answer to question 4: edge and depth were essentially fixed by failure modes (shape collapse and depth flattening), and the real tuning space was the vis–seg balance within a narrow remaining budget.

5. Number of synthetic clips actually generated, and over how many source cli

   Provided answer to these above.

6. Total real-data duration and capture setup.

   Capture setup was egocentric: person was holding camera recording themselves perform basic household tasks.

The total is 31 minutes and 44 seconds (31:44).

7. Prompt and mask for every shown example (slides 13 to 16).

Cosmos predict which i used for the wrapping the charger variation (slide 15). I

Segmentation masks for the comsos experiments i ran





1. Bucket cleaning → blue countertop (your first and most-run prompt)

"The video captures a person cleaning a solid blue countertop in a well-lit indoor kitchen. The countertop surface is a rich, vivid blue color with a smooth matte finish, clearly visible under the even overhead lighting. The person uses paper towels and a bucket to wipe down the blue countertop surface. The bucket and paper towels retain their original appearance. The walls and surrounding cabinets are neutral colors. The room is lit by bright, even overhead ceiling lights that illuminate the blue countertop uniformly. The camera captures the scene from a close angle, focused on the person's hands and the countertop surface."

2. Shelf clip → blue desk and shelf

"The video captures a person standing at a solid blue painted desk in a well-lit indoor room. The desk surface is a rich, vivid blue color with a smooth matte finish, clearly visible under the even overhead lighting. A matching blue shelf is mounted on the wall above the desk, also painted the same vivid blue. The person holds a white charger cable in both hands and wraps it carefully. The walls of the room are a neutral light color. The room is lit by bright, even overhead ceiling lights that illuminate the blue furniture uniformly. The camera captures the scene from a close angle, focused on the person's hands and the desk surface."

3. Shelf clip → modern white desk with oak shelf

"The video captures a person standing at a clean, modern desk in a well-lit contemporary indoor room. The desk has a smooth matte white laminate surface with minimalist straight edges and slim tapered legs. A sleek modern floating shelf is mounted on the wall above the desk, made of light natural oak wood with a clear satin finish, clean straight lines, and concealed mounting hardware giving it a seamless floating appearance. The person holds a white charger cable in both hands and wraps it carefully. The walls are painted a soft warm white. The room is lit by bright, even overhead ceiling lights that cast a clean, shadow-free illumination across the furniture surfaces. The camera captures the scene from a close angle, focused on the person's hands and the desk surface."

4. Pillow casing → blue blanket

"The video captures a person folding a white cloth on a neatly made bed in a well-lit bedroom. A folded blanket rests on the left side of the bed, made of soft fabric in a rich, vivid cobalt blue color with a smooth matte finish. The blue blanket is neatly folded and sits prominently on the white bedding. The white cloth being folded, the decorative pillow with yellow and black patterns, the dark wooden headboard, and the brass decorative item all retain their original colors and appearance. The room is lit by bright, even indoor lighting."

5. Pillow casing → chunky knit texture blanket (partial — I generated this but didn’t find the full text in the search results; it followed the same structure as #4 but with a “chunky cream knit” texture description instead of cobalt blue)

6. Fixing bed → navy blue area rug (the spec you just showed me)

"The video captures a person making a bed by smoothing a beige bedspread in a bright bedroom. On the floor beside the bed is a simple solid navy blue area rug with a flat woven texture and clean straight edges. The

bedspread, the person's hands, and the bed retain their original appearance.
The room is lit by bright indoor lighting."

8. Whether any domain-randomization baseline exists or can be run.

Im not sure how we would run this and if a baseline exists

9. GPU wall-clock per clip.

Experiments run on google colab

Bounding box + prompt generation = `129.5s` (cosmos nano 3 reason) = A100

Cosmos transfer time on 20 minutes per clip on A100 80GB (Colab Pro+).

Cosmos predict: 3.5 min per clip

10. Clearance to publish clips, prompts, and the ontology schema.

We have clearance to publish everything (i asked frank and he said he wanted us to publish pictures from the clips.

Cosmos reason prompt

```
# ── Prompt ──────────────────────────────────────────────────────────────────


COSMOS3_BBOX_VARIATION_PROMPT = """You are an expert robotics training
data analyst with precise spatial grounding.
```

```
Analyze this egocentric image of a person performing a household task.

Your goals:
1. Identify the most task-relevant objects in the scene (up to 10).
   Only include objects that are directly involved in or essential to
   understanding the task.
2. Provide a precise 2D bounding box for each object
3. Enumerate EVERY possible variation for synthetic data generation

BOUNDING BOX FORMAT:
For each object, provide "bbox_2d": [x1, y1, x2, y2] using normalized
coordinates in the 0-1000 range (top-left corner is [0,0], bottom-right
is [1000,1000]). Be precise — the box should tightly enclose the object.

OUTPUT FORMAT — respond in this exact JSON:

{
  "location": {
    "room_type": "",
    "setting": "",
    "time_of_day": "",
    "lighting_condition": "",
    "room_style": ""
```

```
},


"task": {
  "name": "",
  "description": "",
  "phase": "",
  "action_primitive": "",
  "difficulty": "",
  "typical_duration_seconds": 0
},


"key_objects": [
  {
    "rank": 1,
    "name": "",
    "bbox_2d": [0, 0, 0, 0],
    "is_primary": true,
    "role_in_task": "",


    "current_appearance": {
      "color_description": "",
      "rgb": [0.0, 0.0, 0.0],
      "hex": "",
```

```
    "material": "",
    "texture": "",
    "condition": "",
    "size_estimate": ""
  },

  "current_state": {
    "position_on_surface": "",
    "height_above_surface": "",
    "orientation": "",
    "interaction_state": "",
    "task_completion_phase": ""
  },

  "variations": {
    "color_and_appearance": [],
    "material_and_texture": [],
    "size_and_shape": [],
    "state_and_position": [],
    "condition": [],
    "interaction": [],
    "brand_and_style": []
  },
```

```
    "robustness_gaps": {
      "what_robot_must_handle": [],
      "common_failure_modes": [],
      "edge_cases": []
    }
  }
],

"scene_variations": {
  "lighting": [],
  "camera_viewpoint": [],
  "room_style_and_culture": [],
  "time_of_day": [],
  "background_clutter": [],
  "person_characteristics": [],
  "task_sequence_variations": []
},

"dataset_gaps": {
  "what_current_clip_shows": "",
  "what_is_missing": [],
  "highest_priority_variations": []
```

```
  }
}

INSTRUCTIONS:

BOUNDING BOXES: For EVERY object in key_objects, the "bbox_2d" field is
MANDATORY. Coordinates are [x1, y1, x2, y2] normalized to 0-1000 range.
x1,y1 is the top-left corner. x2,y2 is the bottom-right corner. Be tight
and precise. Objects partially out of frame should have boxes clipped to
the frame boundary (0 or 1000).

LOCATION: Be specific — not just "bedroom" but "small cluttered student
bedroom with dark carpet".

TASK: Identify the exact action primitive. "Making bed" is too broad —
"pulling bedspread from foot toward headboard with both hands" is correct.

KEY OBJECTS: For each, enumerate EVERY variation dimension exhaustively:

  color_and_appearance: every color family, pattern, print. Include 10+
    solid colors, patterns (striped, plaid, floral, geometric, plain),
    finishes (matte, glossy, metallic), worn/faded/stained versions.
```

```
  material_and_texture: every realistic material. Include tactile
    properties relevant to robot grasping.

  size_and_shape: every realistic size variation. Different aspect
    ratios. Folded vs unfolded. Stacked vs single.

  state_and_position: every position/orientation/arrangement during
    this task. Before/during/after each action step. Every failure state.

  condition: new, used, worn, dirty, wet, damaged, wrinkled, stained,
    torn, faded, dusty, sticky.

  interaction: how a robot gripper might interact. Grasping points,
    approach angles, grip types, force, fragility, slipperiness.

  brand_and_style: realistic brand/style variations that change
    appearance without changing function.

SCENE VARIATIONS: Be exhaustive for lighting, camera viewpoint,
room style, time of day, background clutter, person characteristics,
and task sequence variations.

DATASET GAPS: What does training ONLY on this clip fail to capture?
```

```
Rank by impact on robot policy robustness.



Answer the question in the following format:

<think>

your reasoning

</think>



<answer>

your JSON here

</answer>"""
```